

AR(2) Process - Real, Distinct Roots

$$\frac{1}{(1-\lambda_1 B)(1-\lambda_2 B)} = \frac{A}{1-\lambda_1 B} + \frac{C}{1-\lambda_2 B}$$

$$\begin{aligned} \Rightarrow 1 &= A(1-\lambda_2 B) + C(1-\lambda_1 B) \\ &= A + C - (\lambda_2 A + \lambda_1 C)B \end{aligned}$$

Equate coefficients:

$$A + C = 1 \quad (1)$$

$$\lambda_2 A + \lambda_1 C = 0 \quad (2)$$

$$\text{From (1)} \quad C = 1 - A$$

Substitute the above into (2):

$$\Rightarrow \lambda_2 A + \lambda_1 (1 - A) = 0$$

$$A(\lambda_2 - \lambda_1) = -\lambda_1$$

$$\Rightarrow A = \frac{-\lambda_1}{\lambda_2 - \lambda_1} = \frac{\lambda_1}{\lambda_1 - \lambda_2}$$

$$C = 1 - A = 1 - \frac{\lambda_1}{\lambda_1 - \lambda_2}$$

$$= \frac{-\lambda_2}{\lambda_1 - \lambda_2}$$

$$\therefore \Psi(B) = \frac{\lambda_1}{\lambda_1 - \lambda_2} \left[\frac{1}{1 - \lambda_1 B} \right] - \frac{\lambda_2}{\lambda_1 - \lambda_2} \left[\frac{1}{1 - \lambda_2 B} \right]$$

$$X_t = \Psi(B) Z_t$$

$$\frac{1}{1 - \lambda_1 B} = \sum_{j=0}^{\infty} \lambda_1^j B^j$$

$$\frac{1}{1 - \lambda_2 B} = \sum_{j=0}^{\infty} \lambda_2^j B^j$$

Taylor series expansions
Lecture 2.

$$X_t = \left\{ \frac{\lambda_1}{\lambda_1 - \lambda_2} \sum_{j=0}^{\infty} \lambda_1^j B^j - \frac{\lambda_2}{\lambda_1 - \lambda_2} \sum_{j=0}^{\infty} \lambda_2^j B^j \right\} Z_t$$

$\Psi(B)$

$$= \frac{1}{\lambda_1 - \lambda_2} \left[\sum_{j=0}^{\infty} \lambda_1^{j+1} B^j - \sum_{j=0}^{\infty} \lambda_2^{j+1} B^j \right] Z_t$$

$$= \frac{1}{\lambda_1 - \lambda_2} \sum_{j=0}^{\infty} (\lambda_1^{j+1} - \lambda_2^{j+1}) Z_{t-j}$$

$$= \sum_{j=0}^{\infty} \beta_j Z_{t-j} \quad \beta_j = \frac{\lambda_1^{j+1} - \lambda_2^{j+1}}{\lambda_1 - \lambda_2}$$